ASSIGNMENT -01

APACHE HIVE

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Scenario

A global retail company stores transaction records in flat CSV files on HDFS. The business intelligence team requires structured SQL access to analyze revenue patterns across store categories.

Problem Statement

Create an external table in Hive mapping to raw transaction data stored on HDFS. Write HiveQL queries to clean missing data, aggregate total sales revenue, and calculate the average order value per product category.

Apache Hive & Hadoop

Apache Hive is a data warehouse and data analysis tool built on Hadoop. It provides a SQL-like language called HiveQL to query and analyze data stored in HDFS.

Apache Hadoop is an open-source framework used for distributed storage and processing of large datasets. It uses components such as HDFS, MapReduce, and YARN to manage and process big data.

Docker

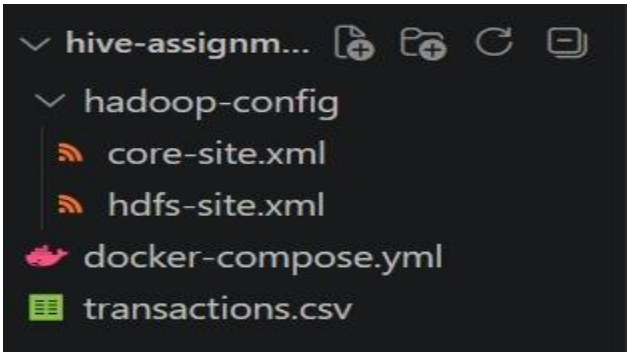
Docker is used to provide a consistent and isolated environment for running applications such as Hadoop and Apache Hive. In our project, Docker can help run Hive in a Linux-based container because the Apache Hive installation we have uses Linux shell scripts rather than Windows .cmd files. This avoids many Windows configuration and compatibility problems and makes it easier to set up Hadoop, HDFS, and Hive together. Thus, Hadoop provides storage and processing, Hive performs SQL-like data analysis, and Docker provides the environment to run these tools smoothly.

Workflow

Docker Desktop → Hadoop/HDFS → Transaction CSV → Apache Hive → External Table → Data Cleaning → HiveQL Aggregation → Revenue & Average Order Value.

Docker-Compose.yml

The Docker Compose workflow uses **Docker Desktop** to run a Hadoop and Apache Hive environment through separate containers. The **NameNode** manages the HDFS file system, while the **DataNode** stores the actual transaction data. Apache Hive runs through the HiveServer2 container and provides HiveQL for querying the data stored in HDFS. The transaction CSV file is uploaded to HDFS, an external Hive table is created over the data, and HiveQL queries are used to clean missing records, calculate total sales revenue, and determine the average order value for each product category.



docker-compose.yml

```yaml
services:

  namenode:
    image: apache/hadoop:3.4.1
    container_name: hive-namenode
    hostname: namenode
    command: >
      bash -c "
      if [ ! -d /data/name/current ]; then
        hdfs namenode -format -force;
      fi;
      hdfs namenode
      "
    environment:
      HADOOP_HOME: /opt/hadoop
      HADOOP_CONF_DIR: /opt/hadoop/etc/hadoop
    volumes:
      - namenode_data:/data
      - ./hadoop-config/core-site.xml:/opt/hadoop/etc/hadoop/core-site.xml
      - ./hadoop-config/hdfs-site.xml:/opt/hadoop/etc/hadoop/hdfs-site.xml
    ports:
      - "9870:9870"
      - "9000:9000"


  datanode:
    image: apache/hadoop:3.4.1
    container_name: hive-datanode
    hostname: datanode
    depends_on:
      - namenode
    command: ["hdfs", "datanode"]
    environment:
      HADOOP_HOME: /opt/hadoop
      HADOOP_CONF_DIR: /opt/hadoop/etc/hadoop
    volumes:
      - datanode_data:/data
      - ./hadoop-config/core-site.xml:/opt/hadoop/etc/hadoop/core-site.xml
      - ./hadoop-config/hdfs-site.xml:/opt/hadoop/etc/hadoop/hdfs-site.xml

  hive:
    image: apache/hive:4.0.1
    container_name: hive-server
    hostname: hive-server
    depends_on:
      - namenode
      - datanode
    environment:
      SERVICE_NAME: hiveserver2
    ports:
      - "10000:10000"
      - "10002:10002"
volumes:
  namenode_data:
  datanode_data:
```

Transaction.csv

transactions.csv

```
transaction_id,product_name,category,quantity,price
T001,Laptop,Electronics,2,55000
T002,Mouse,Electronics,5,800
T003,Office Chair,Furniture,2,7500
T004,Notebook,Stationery,10,120
T005,Keyboard,Electronics,3,1500
T006,Desk,Furniture,1,12000
T007,Pen,Stationery,20,50
T008,Monitor,Electronics,2,18000
T009,Bookshelf,Furniture,1,9000
T010,Printer,,1,15000
T011,Headphones,Electronics,,2500
T012,Paper,Stationery,5,300
T013,Table,Furniture,2,6500
T014,Webcam,Electronics,2,2200
T015,Marker,Stationery,10,80
```

1.To check docker status

```
D:\hive-assignment>docker compose ps
NAME            IMAGE                COMMAND                  SERVICE    CREATED       STATUS         PORTS
hive-datanode   apache/hadoop:3.4.1   "/usr/local/bin/dumb…"   datanode   9 hours ago   Up 5 seconds
hive-namenode   apache/hadoop:3.4.1   "/usr/local/bin/dumb…"   namenode   9 hours ago   Up 5 seconds   0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp, 0.0.0.0:
9870->9870/tcp, [::]:9870->9870/tcp
hive-server     apache/hive:4.0.1     "sh -c /entrypoint.sh"   hive       9 hours ago   Up 4 seconds   0.0.0.0:10000->10000/tcp, [::]:10000->10000/tcp, 0.0.
0.0:10002->10002/tcp, [::]:10002->10002/tcp
```

2.Command Line Interface (CLI)

```
D:\hive-assignment>docker exec -it hive-server beeline -u jdbc:hive2://localhost:10000
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/opt/hive/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/opt/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/opt/hive/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/opt/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Connecting to jdbc:hive2://localhost:10000
Connected to: Apache Hive (version 4.0.1)
Driver: Hive JDBC (version 4.0.1)
Transaction isolation: TRANSACTION_REPEATABLE_READ
Beeline version 4.0.1 by Apache Hive
0: jdbc:hive2://localhost:10000>
```

3.To create new database

```
0: jdbc:hive2://localhost:10000> CREATE DATABASE retail_db;
INFO  : Compiling command(queryId=hive_20260925152520_0e262785-5ce6-49f6-a4ac-47a28d8cb1ac): CREATE DATABASE retail_db
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:null, properties:null)
INFO  : Completed compiling command(queryId=hive_20260925152520_0e262785-5ce6-49f6-a4ac-47a28d8cb1ac); Time taken: 0.186 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260925152520_0e262785-5ce6-49f6-a4ac-47a28d8cb1ac): CREATE DATABASE retail_db
INFO  : Starting task [Stage-0:DDL] in serial mode
INFO  : Completed executing command(queryId=hive_20260925152520_0e262785-5ce6-49f6-a4ac-47a28d8cb1ac); Time taken: 0.052 seconds
No rows affected (0.309 seconds)
```

4.**To create External table**

```
0: jdbc:hive2://localhost:10000> CREATE EXTERNAL TABLE transactions (
. . . . . .>     transaction_id STRING,
. . . . . .>     product_name STRING,
. . . . . .>     category STRING,
. . . . . .>     quantity INT,
. . . . . .>     price DOUBLE
. . . . . .> )
. . . . . .> ROW FORMAT DELIMITED
. . . . . .> FIELDS TERMINATED BY ','
. . . . . .> STORED AS TEXTFILE
. . . . . .> LOCATION 'hdfs://namenode:9000/retail_data'
. . . . . .> TBLPROPERTIES ("skip.header.line.count"="1");
```

```
0: jdbc:hive2://localhost:10000> SELECT * FROM transactions;
INFO  : Compiling command(queryId=hive_20260925152830_9a815af4-5926-473b-9710-aaa5d0edbfb4): SELECT * FROM transactions
INFO  : No Stats for retail_db@transactions, Columns: transaction_id, quantity, price, category, product_name
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:transactions.transaction_id, type:string, comment:null), FieldSchema(name:tr
oduct_name, type:string, comment:null), FieldSchema(name:transactions.category, type:string, comment:null), FieldSchema(name:transactions.quant
, comment:null), FieldSchema(name:transactions.price, type:double, comment:null)], properties:null)
INFO  : Completed compiling command(queryId=hive_20260925152830_9a815af4-5926-473b-9710-aaa5d0edbfb4); Time taken: 2.35 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260925152830_9a815af4-5926-473b-9710-aaa5d0edbfb4): SELECT * FROM transactions
INFO  : Completed executing command(queryId=hive_20260925152830_9a815af4-5926-473b-9710-aaa5d0edbfb4); Time taken: 0.001 seconds
+-----------------------------+-------------------------------+-------------------------+-------------------------+-----------------------+
| transactions.transaction_id | transactions.product_name     | transactions.category   | transactions.quantity   | transactions.price    |
+-----------------------------+-------------------------------+-------------------------+-------------------------+-----------------------+
| T001                        | Laptop                        | Electronics             | 2                       | 55000.0               |
| T002                        | Mouse                         | Electronics             | 5                       | 800.0                 |
| T003                        | Office Chair                  | Furniture               | 2                       | 7500.0                |
| T004                        | Notebook                      | Stationery              | 10                      | 120.0                 |
| T005                        | Keyboard                      | Electronics             | 3                       | 1500.0                |
| T006                        | Desk                          | Furniture               | 1                       | 12000.0               |
| T007                        | Pen                           | Stationery              | 20                      | 50.0                  |
| T008                        | Monitor                       | Electronics             | 2                       | 18000.0               |
| T009                        | Bookshelf                     | Furniture               | 1                       | 9000.0                |
| T010                        | Printer                       |                         | 1                       | 15000.0               |
| T011                        | Headphones                    | Electronics             | NULL                    | 2500.0                |
| T012                        | Paper                         | Stationery              | 5                       | 300.0                 |
| T013                        | Table                         | Furniture               | 2                       | 6500.0                |
| T014                        | Webcam                        | Electronics             | 2                       | 2200.0                |
| T015                        | Marker                        | Stationery              | 10                      | 80.0                  |
+-----------------------------+-------------------------------+-------------------------+-------------------------+-----------------------+
15 rows selected (2.811 seconds)
```

5.**Identifying the missing values and cleaning the table**

```
0: jdbc:hive2://localhost:10000> CREATE TABLE cleaned_transactions AS
. . . . . .> SELECT
. . . . . .>     transaction_id,
. . . . . .>     product_name,
. . . . . .>     CASE
. . . . . .>         WHEN category IS NULL OR TRIM(category) = '' THEN 'Unknown'
. . . . . .>         ELSE category
. . . . . .>     END AS category,
. . . . . .>     CASE
. . . . . .>         WHEN quantity IS NULL THEN 1
. . . . . .>         ELSE quantity
. . . . . .>     END AS quantity,
. . . . . .>     price
. . . . . .> FROM transactions;
```

```
0: jdbc:hive2://localhost:10000> SELECT *
. . . . . .> FROM transactions
. . . . . .> WHERE category IS NULL
. . . . . .>    OR TRIM(category) = ''
. . . . . .>    OR quantity IS NULL;
INFO  : Compiling command(queryId=hive_20260925152851_4bbbaa32-aafc-4e0e-82a1-75026b4ef821): SELECT *
FROM transactions
WHERE category IS NULL
OR TRIM(category) = ''
OR quantity IS NULL
INFO  : No Stats for retail_db@transactions, Columns: transaction_id, quantity, price, category, product_name
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:transactions.transaction_id, type:string, comment:null), FieldSchema
oduct_name, type:string, comment:null), FieldSchema(name:transactions.category, type:string, comment:null), FieldSchema(name:transactio
, comment:null), FieldSchema(name:transactions.price, type:double, comment:null)], properties:null)
INFO  : Completed compiling command(queryId=hive_20260925152851_4bbbaa32-aafc-4e0e-82a1-75026b4ef821); Time taken: 0.505 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260925152851_4bbbaa32-aafc-4e0e-82a1-75026b4ef821): SELECT *
FROM transactions
WHERE category IS NULL
OR TRIM(category) = ''
OR quantity IS NULL
INFO  : Completed executing command(queryId=hive_20260925152851_4bbbaa32-aafc-4e0e-82a1-75026b4ef821); Time taken: 0.001 seconds
+-----------------------------+-------------------------------+-------------------------+-------------------------+-----------------------+
| transactions.transaction_id | transactions.product_name     | transactions.category   | transactions.quantity   | transactions.price    |
+-----------------------------+-------------------------------+-------------------------+-------------------------+-----------------------+
| T010                        | Printer                       |                         | 1                       | 15000.0               |
| T011                        | Headphones                    | Electronics             | NULL                    | 2500.0                |
+-----------------------------+-------------------------------+-------------------------+-------------------------+-----------------------+
2 rows selected (0.562 seconds)
```

```
0: jdbc:hive2://localhost:10000> SELECT * FROM cleaned_transactions;
INFO  : Compiling command(queryId=hive_20260925153022_0033bd91-6f25-4a83-aff8-9025a86a763a): SELECT * FROM cleaned_transactions
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:cleaned_transactions.transaction_id, type:string, comment:null), FieldSchema(name:cleaned_transactions.produc
:string, comment:null), FieldSchema(name:cleaned_transactions.category, type:string, comment:null), FieldSchema(name:cleaned_transactions.quantity, type:int, comment:null), Fie
e:cleaned_transactions.price, type:double, comment:null)], properties:null)
INFO  : Completed compiling command(queryId=hive_20260925153022_0033bd91-6f25-4a83-aff8-9025a86a763a); Time taken: 0.1 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260925153022_0033bd91-6f25-4a83-aff8-9025a86a763a): SELECT * FROM cleaned_transactions
INFO  : Completed executing command(queryId=hive_20260925153022_0033bd91-6f25-4a83-aff8-9025a86a763a); Time taken: 0.0 seconds
+-----------------------------------+-------------------------------+-----------------------------+--------------------------------+-----------------------------+
| cleaned_transactions.transaction_id | cleaned_transactions.product_name | cleaned_transactions.category | cleaned_transactions.quantity | cleaned_transactions.price |
+-----------------------------------+-------------------------------+-----------------------------+--------------------------------+-----------------------------+
| T001                              | Laptop                        | Electronics                 | 2                              | 55000.0                     |
| T002                              | Mouse                         | Electronics                 | 5                              | 800.0                       |
| T003                              | Office Chair                  | Furniture                   | 2                              | 7500.0                      |
| T004                              | Notebook                      | Stationery                  | 10                             | 120.0                       |
| T005                              | Keyboard                      | Electronics                 | 3                              | 1500.0                      |
| T006                              | Desk                          | Furniture                   | 1                              | 12000.0                     |
| T007                              | Pen                           | Stationery                  | 20                             | 50.0                        |
| T008                              | Monitor                       | Electronics                 | 2                              | 18000.0                     |
| T009                              | Bookshelf                     | Furniture                   | 1                              | 9000.0                      |
| T010                              | Printer                       | Unknown                     | 1                              | 15000.0                     |
| T011                              | Headphones                    | Electronics                 | 1                              | 2500.0                      |
| T012                              | Paper                         | Stationery                  | 5                              | 300.0                       |
| T013                              | Table                         | Furniture                   | 2                              | 6500.0                      |
| T014                              | Webcam                        | Electronics                 | 2                              | 2200.0                      |
| T015                              | Marker                        | Stationery                  | 10                             | 80.0                        |
+-----------------------------------+-------------------------------+-----------------------------+--------------------------------+-----------------------------+
15 rows selected (0.129 seconds)
```

6. **Calculate total sales revenue**

```
0: jdbc:hive2://localhost:10000> SELECT
. . . . . . . . . . . . . . . .>     SUM(quantity * price) AS total_sales_revenue
. . . . . . . . . . . . . . . .> FROM cleaned_transactions;
INFO  : Compiling command(queryId=hive_20260925153052_979e6c9c-9136-4f6b-9790-9813e
SUM(quantity * price) AS total_sales_revenue
FROM cleaned_transactions
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:total_sales_reve
INFO  : Completed compiling command(queryId=hive_20260925153052_979e6c9c-9136-4f6b-
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260925153052_979e6c9c-9136-4f6b-9790-9813e
SUM(quantity * price) AS total_sales_revenue
FROM cleaned_transactions
INFO  : Query ID = hive_20260925153052_979e6c9c-9136-4f6b-9790-9813e5011c4b
INFO  : Total jobs = 1
INFO  : Launching Job 1 out of 1
INFO  : Starting task [Stage-1:MAPRED] in serial mode
INFO  : Subscribed to counters: [] for queryId: hive_20260925153052_979e6c9c-9136-4
INFO  : Session is already open
INFO  : Dag name: SELECT
SUM(quantity *......aned_transactions (Stage-1)
INFO  : HS2 Host: [hive-server], Query ID: [hive_20260925153052_979e6c9c-9136-4f6b-
INFO  : Status: Running (Executing on YARN cluster with App id application_17903501

INFO  : Completed executing command(queryId=hive_20260925153052_979e6c9c-9136-4f6b-
+----------------------+
| total_sales_revenue  |
+----------------------+
| 229900.0             |
+----------------------+
1 row selected (0.886 seconds)
```

7.**Calculate average order value by category**

```
0: jdbc:hive2://localhost:10000> SELECT
. . . . . .>     category,
. . . . . .>     AVG(quantity * price) AS average_order_value
. . . . . .> FROM cleaned_transactions
. . . . . .> GROUP BY category;
INFO  : Compiling command(queryId=hive_20260925153123_bdac5e29-0157-4ad9-8cef-156382e2046b): SELECT
category,
AVG(quantity * price) AS average_order_value
FROM cleaned_transactions
GROUP BY category
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:category, type:string, comment:null), FieldSch
INFO  : Completed compiling command(queryId=hive_20260925153123_bdac5e29-0157-4ad9-8cef-156382e2046b); Time taken
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260925153123_bdac5e29-0157-4ad9-8cef-156382e2046b): SELECT
category,
AVG(quantity * price) AS average_order_value
FROM cleaned_transactions
GROUP BY category
INFO  : Query ID = hive_20260925153123_bdac5e29-0157-4ad9-8cef-156382e2046b
INFO  : Total jobs = 1
INFO  : Launching Job 1 out of 1
INFO  : Starting task [Stage-1:MAPRED] in serial mode
INFO  : Subscribed to counters: [] for queryId: hive_20260925153123_bdac5e29-0157-4ad9-8cef-156382e2046b
INFO  : Session is already open
INFO  : Dag name: SELECT
category,
AVG(......GROUP BY category (Stage-1)
INFO  : HS2 Host: [hive-server], Query ID: [hive_20260925153123_bdac5e29-0157-4ad9-8cef-156382e2046b], Dag ID: [da
INFO  : Status: Running (Executing on YARN cluster with App id application_1790350157453_0001)

INFO  : Completed executing command(queryId=hive_20260925153123_bdac5e29-0157-4ad9-8cef-156382e2046b); Time taken
+--------------+----------------------+
|   category   | average_order_value  |
+--------------+----------------------+
| Electronics  | 26900.0              |
| Furniture    | 12250.0              |
| Stationery   | 1125.0               |
| Unknown      | 15000.0              |
+--------------+----------------------+
4 rows selected (1.009 seconds)
```

Conclusion

This assignment successfully demonstrates the use of Apache Hadoop, HDFS, Apache Hive, and Docker for data engineering. Transaction data is stored in HDFS and analyzed using an external Hive table and HiveQL queries. Missing data is cleaned, and total sales revenue and average order value are calculated for each product category. Docker provides a consistent environment for running the required technologies, making the overall data analysis workflow easier to manage.